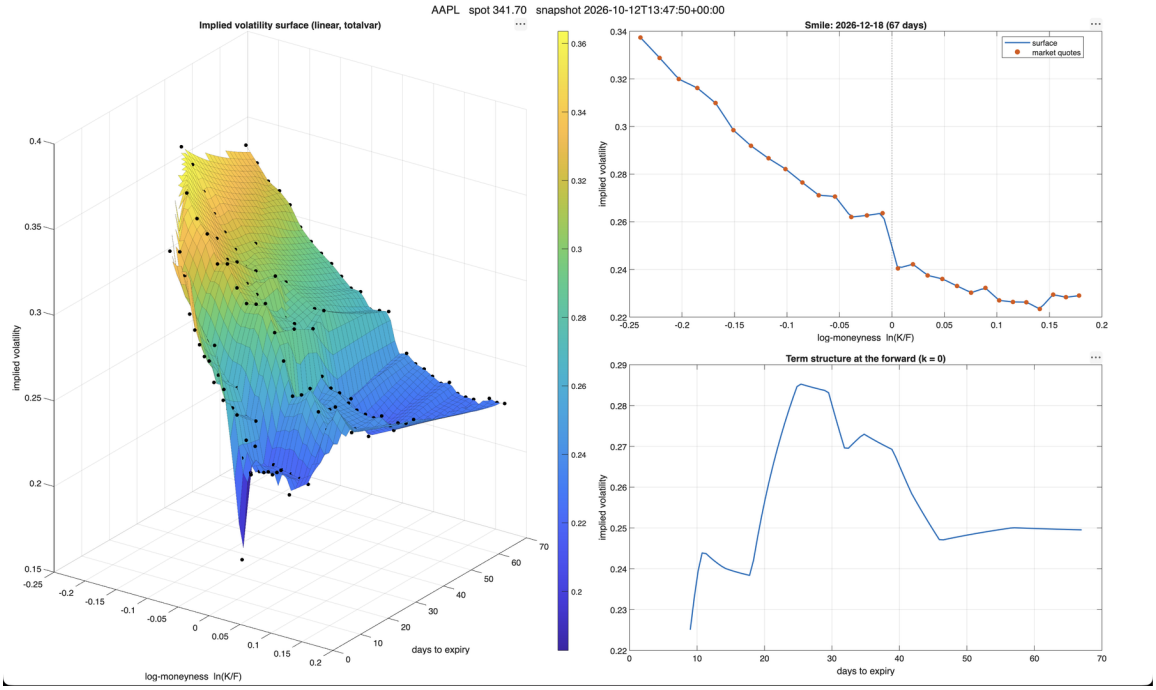
Volatility Surface Tool — User Guide

Type one ticker, get an interactive implied-volatility surface

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1 What this tool does

`vol_surface("AAPL")` downloads the current option chain for a stock, works out the *implied volatility* of every liquid option, and draws three linked views: a 3-D surface, the "smile" for one expiry, and the term structure. A second window explains how to read all of it in plain language. Under the hood it connects a Python/OCaml option-pricing engine to MATLAB's symbolic and numerical tools.



2 Before you start

Need	Details
MATLAB	R2020a or newer, with the Symbolic Math Toolbox and the Statistics and Machine Learning Toolbox
Python	A Python with `yfinance`, `pandas`, `numpy` and `scipy`. The tool looks in `<repo>/.venv`, then MATLAB's configured Python, then `python3`
OCaml engine	Optional. Adds a second, independent volatility for cross-checking. Turn off with `"UseOCaml", false`
Internet	Needed to fetch a chain from Yahoo Finance

3 Quick start

1. In MATLAB, set the current folder to the repository's `matlab` folder.
2. Run one line:

```
vol_surface("AAPL")
```

3. Wait 10–60 seconds for the download. Two windows open: the surface figure and *How to read the surface*.

Ticker symbols may contain letters, digits, "." and "-" (up to 10 characters). The tool writes a dated snapshot to `snapshots/` so you can reuse it later.

4 When to run it

Yahoo Finance serves only the *current* chain, and it is meaningful when the market is open (9:30–16:00 Eastern, weekdays). Before the open most bids and asks are zero.

Situation	Command
Market open (best)	`vol_surface("AAPL")`
Too early in the day	`vol_surface("AAPL", "Session", "prior")` rebuilds yesterday's close from each option's last trade
Not sure	`vol_surface("AAPL", "Session", "auto")` tries live, then falls back to the prior session
Offline or repeat a run	`vol_surface("AAPL", "Refresh", false)` reuses the saved snapshot

Prior-session mode keeps only trades from the last `PriorWindowMin` minutes (default 60) of the previous session and prices them against that session's closing price. It also works shortly after the open: the tool measures the share of contracts that have already traded today and refuses if that exceeds 20%, because liquid contracts trade first and the sample would skew toward illiquid strikes. Prior surfaces are noisier than live ones, and the figure title says so.

5 Options

Name	Default	Meaning
`Refresh`	`true`	Fetch live data; false uses the cached snapshot. If a fetch fails and a cache exists, the cache is used with a warning
`Session`	`"live"`	"live", "prior" or "auto" (see above)
`PriorWindowMin`	`60`	Prior mode: keep trades from the last N minutes of that session
`MaxExpiries`	`8`	Number of expiries to download
`MinDays`	`7`	Skip expiries closer than this many days
`Expiry`	`""`	Expiry shown in the smile panel, e.g. "2026-11-20". Default: the expiry with the most quotes
`UseOCaml`	`true`	Add the OCaml reference volatility and cross-check against it
`Python`	`""`	Path to a specific Python. Default: automatic search
`Explain`	`true`	Open the plain-language explainer window

Example combining several options:

```
out = vol_surface("MSFT", "MaxExpiries", 6, "Expiry", "2026-11-20", "Explain
    ", false);
```

6 Reading the results

- **Surface (left).** Height is implied volatility. The horizontal axes are how far the strike is from the current price (log-moneyness, 0 = at the forward) and days to expiry. Black dots are real quotes. Blank areas have no reliable data and are never extrapolated.
- **Smile (top right).** One expiry cut out of the surface, with the market quotes as dots.
- **Term structure (bottom right).** Volatility at the forward price across expiries.
- **Explainer window.** Six panels that define options, volatility, implied volatility, the smile and the term structure using this ticker's numbers.

The optional return value `out` is a struct with `csv`, `chain`, `res`, `surface`, `smile`, `term`, `figure` and `explainFigure`, handy for your own analysis.

7 Using the pieces separately

Any chain CSV works, not only the tool's own snapshots.

```
res = run_vol_checks("snapshots/AAPL_latest.csv");   % solve + compare to
    OCaml
s   = build_surface(res);                            % interpolated surface
sl  = smile_slice(s, res, "2026-11-20");             % one expiry
ts  = term_structure_slice(s, 0);                    % at-the-forward term
    structure
```

Required CSV columns (any case): `spot, expiry, T, r, type, strike` and either `mid` or both `bid` and `ask`. Optional: `q` (dividend yield, default 0), `symbol`, and `iv_ocaml` (reference volatility). `T` is years to expiry; `type` is call or put.

8 Checking that it works

```
test_vol_pipeline     % pricing, solver and pipeline, no network needed
test_surface          % interpolation and slices against a known surface
verify_demo           % one-page pass/fail verification report
verify_demo("snapshots/AAPL_latest.csv")   % also compare a real snapshot to
   OCaml
```

The companion document *Verification Process* explains each check.

9 Troubleshooting

Message or symptom	What to do
"has only N usable out-of-the-money quote(s)"	The chain is thin, usually before the open. Rerun when the market is open, use `"Session","prior"`, or pass `"Refresh", false`
"% of contracts have already traded since the prior session closed"	Too much of today's trading has overwritten yesterday's prices. Use live mode
"Python exporter failed"	The Python found lacks `yfinance` or its relatives. Pass `"Python","/path/to/python"`
"No cached snapshot at . . . "	There is nothing saved for this ticker and mode. Run once with `"Refresh", true`
Dip in one short expiry	Few quotes with wide spreads; treat that expiry with caution
Unknown ticker or empty chain	Check the symbol; some tickers have no listed options

10 Limitations

- Black-Scholes with a constant dividend yield treats American single-stock options as European; the tool uses only out-of-the-money quotes to limit the effect.
- The surface is interpolated, not fitted with a no-arbitrage model. Treat it as a faithful picture of the quotes, not a tradable price list.
- Quote data comes from Yahoo Finance and may be delayed or stale.
- Educational use only; nothing here is investment advice.